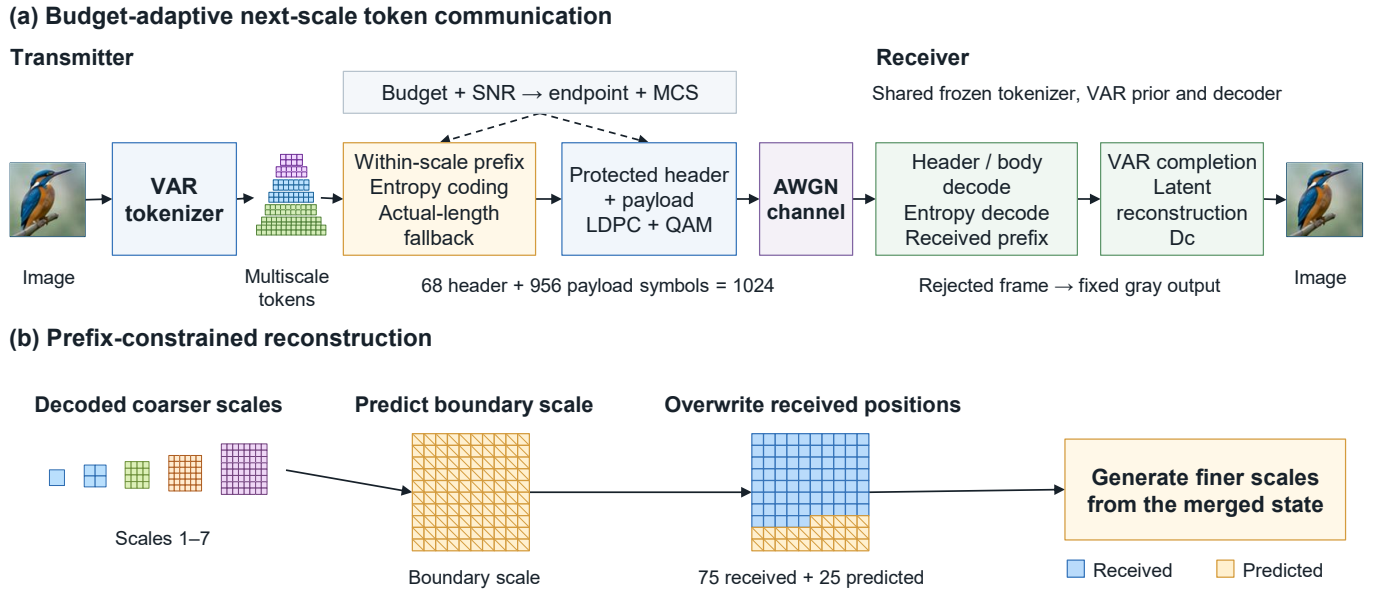

(a) Budget-adaptive next-scale token communication
Transmitter
Receiver
Budget + SNR → endpoint + MCS
Shared frozen tokenizer, VAR prior and decoder
VAR tokenizer
Within-scale prefix Entropy coding Actual-length fallback
Protected header + payload LDPC + QAM
AWGN channel
Header / body decode Entropy decode Received prefix
VAR completion Latent reconstruction Dc
Image
Multiscale tokens
68 header + 956 payload symbols = 1024
Rejected frame → fixed gray output
Image
(b) Prefix-constrained reconstruction
Decoded coarser scales
Predict boundary scale
Overwrite received positions
Generate finer scales from the merged state
Scales 1–7
Boundary scale
75 received + 25 predicted
Received
Predicted